

Some basic game theory

ECN 714

Game theory, General references

- MSZ: Maschler, Solan, and Zamir (2020)
- OR: Osborne and Rubinstein (1994)
- MWG: Mas-Colell, Whinston, and Green (1995)
- M: Myerson (1991)
- FT: Fudenberg and Tirole (1991)

Extensive form

Extensive-form representation

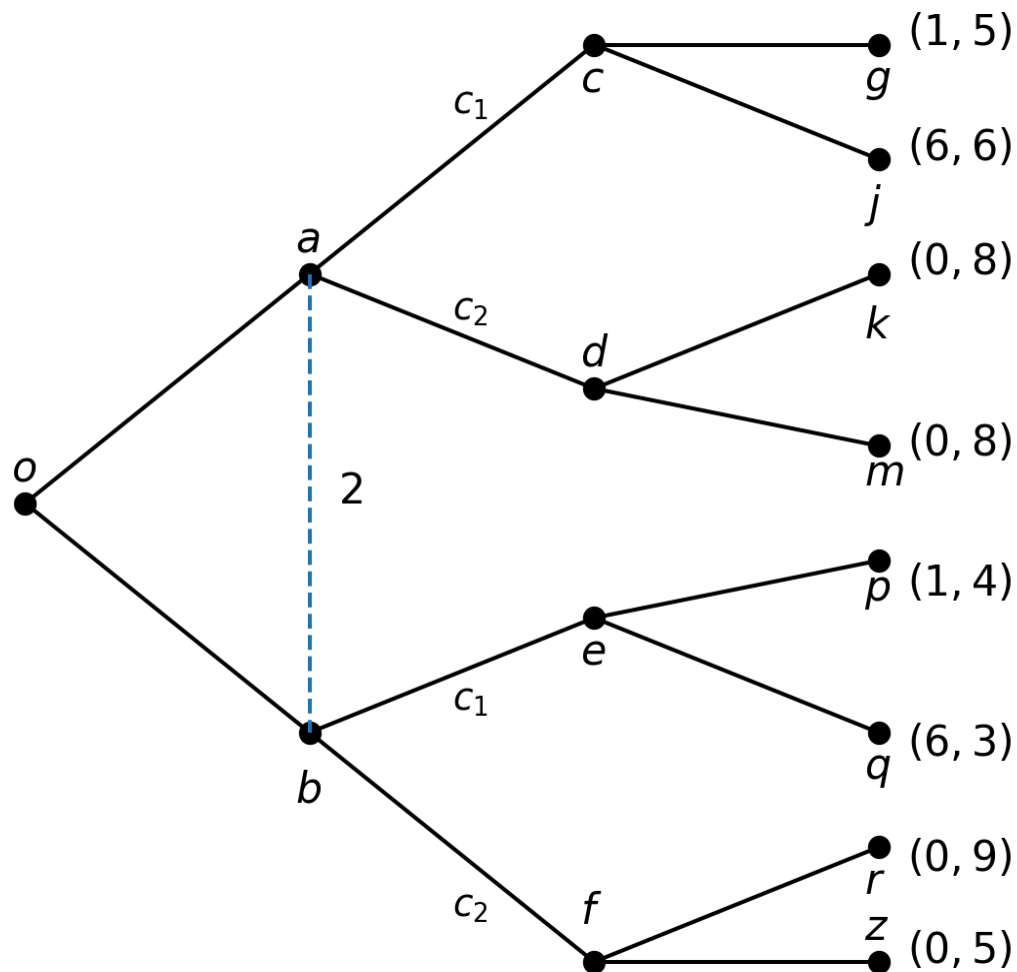
- We will see the definition given by Kuhn (1953). It is an extension of the original formulation by Neumann and Morgenstern (1944). See van Damme (1983) for a readable presentation.

There are various alternative definitions.

- Alternative definition given by Kreps and Wilson (1982).
- Alternative but not equivalent definition Osborne and Rubinstein (1994).
- More general definition, Alós-Ferrer and Ritzberger (2008).

In a simple game without private information or chance nodes: Every node represents the *current state of the game* which includes all information, including the history of moves that lead to that point.

A game in extensive form



$X = \{o, a, b, c, d, e, f, g, j, k, m, p, q, r, z\}$

$\mathcal{P}_1 = \{o, c, d, e, f\}, \mathcal{P}_2 = \{a, b\}$

$\mathcal{U}_1 = \{\{o\}, \{c\}, \{d\}, \{e\}, \{f\}\}$

$\mathcal{U}_2 = \{\{a, b\}\}$

$\mathcal{C}_{\mathcal{U}_2} = \{c_1, c_2\}$ where

$c_1 = \{c, e\}, c_2 = \{d, f\}$

Extensive-form game

Definition 1 (Kuhn) A game in extensive form is a collection

$$\Gamma = [\mathcal{I}, (X, <), \{\mathcal{P}_i\}_{i=0}^I, \{\mathcal{U}_i\}_{i=1}^I, \{\mathcal{C}_h\}_{h \in \mathcal{U}_i, i \geq 1}, \{P_x\}_{x \in \mathcal{P}_0}, V]$$

where the seven elements defining Γ are

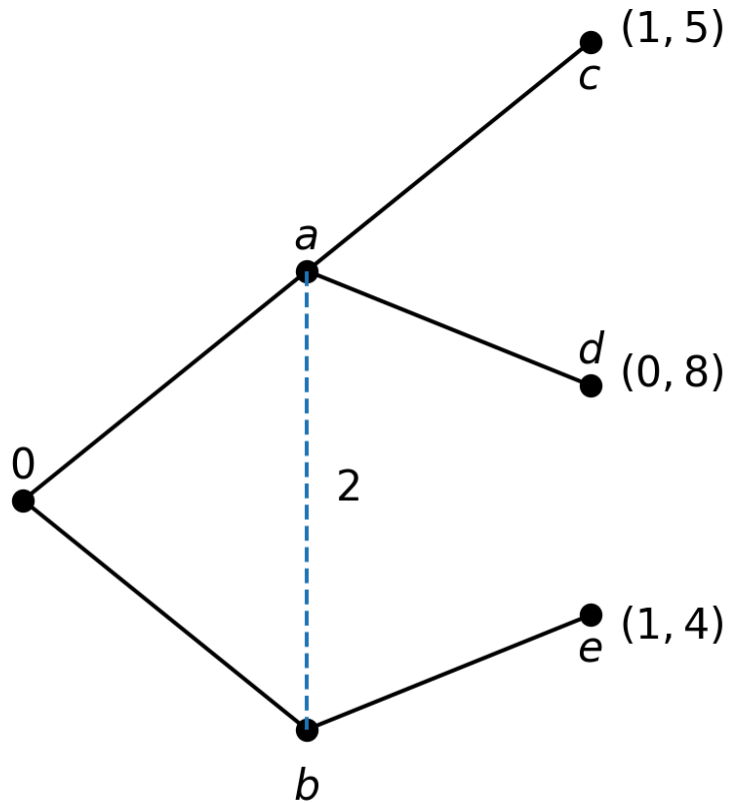
1. $\mathcal{I} = \{0, 1, \dots, I\}$ is a list of players.

Player 0 is interpreted as a *chance player*, responsible for random moves attributed to nature.

2. $(X, <)$ is a game tree:

- X is a collection of nodes and $<$ is an asymmetric, binary relation ($x < y \implies y \not< x$).
- $\exists o \in X$ such that $\forall x \in X, \exists! \{x_1, \dots, x_n\} \subset X$, with $o < x_1 < x_2 < \dots < x_n < x$.
- If $x < y$, x is said to be the immediate predecessor of y .
- $\text{succ}(x) = \{y : y > x\}$ is the set of immediate successors of x .
- $Z = \{x \in X : \nexists y \in X, x < y\}$ is the set of terminal nodes.
- Any $\{x_1, \dots, x_n\} : x_1 < x_2 < \dots < x_n$ is a path from x_1 to x_n .

Example



$X = \{0, a, b, c, d, e\}$

Order: $\{0 < a, 0 < b, a < c, a < d, b < e\}$

$Z = \{c, d, e\}$

3. A partition $\{\mathcal{P}_0, \dots, \mathcal{P}_I\}$ of $X \setminus Z$.

- $\mathcal{P}_i$ is the set of nodes where player i is called to move.

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4. $(\mathcal{U}_1, \mathcal{U}_2, \dots, \mathcal{U}_I)$ where $(\forall i) \mathcal{U}_i$ is a partition of $\mathcal{P}_i : h \in \mathcal{U}_i \implies$
- a. every path intersects h at most once
 - b. $x, y \in h \implies |\text{succ}(x)| = |\text{succ}(y)|$
- An $h \in \mathcal{U}_i$ is an “information set” for player i , i knows that she has reached h but does not know at what node in h she is.
 - 4.a Otherwise a player would not remember whether she has already moved.
 - 4.b Otherwise the player infers the node she is at by the choices she has available.

5. $(\forall i \in \mathcal{I}) (\forall h \in \mathcal{U}_i) \mathcal{C}_h$ is a partition of $\bigcup_{x \in h} \mathbf{succ}(x)$ into choices at h , such that every choice contains exactly one element of $\mathbf{succ}(x)$ for every $x \in h$. This is possible by 4.b.

6. $(\forall x \in \mathcal{P}_0) P_x$ is a completely mixed probability distribution over $\mathbf{succ}(x)$.

If x is reached, $y \in \mathbf{succ}(x)$ is chosen by nature with probability $P_x(y)$.

7. A payoff function, $V : Z \rightarrow \mathbb{R}^I$,

$$V(z) = (V_1(z), \dots, V_I(z)).$$

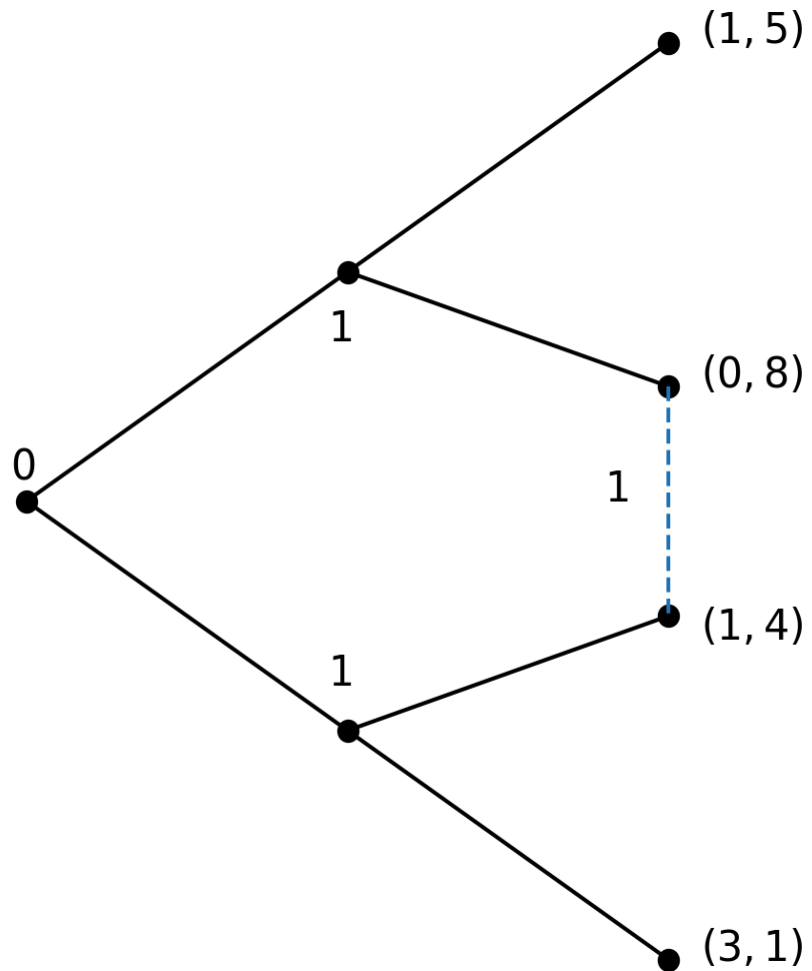
If $z \in Z$ is reached, player i receives payoff $V_i(z)$.

Perfect recall

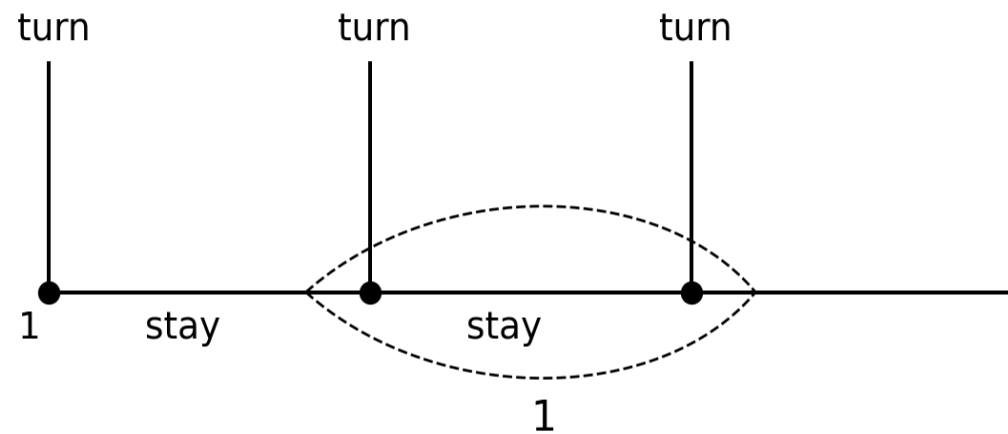
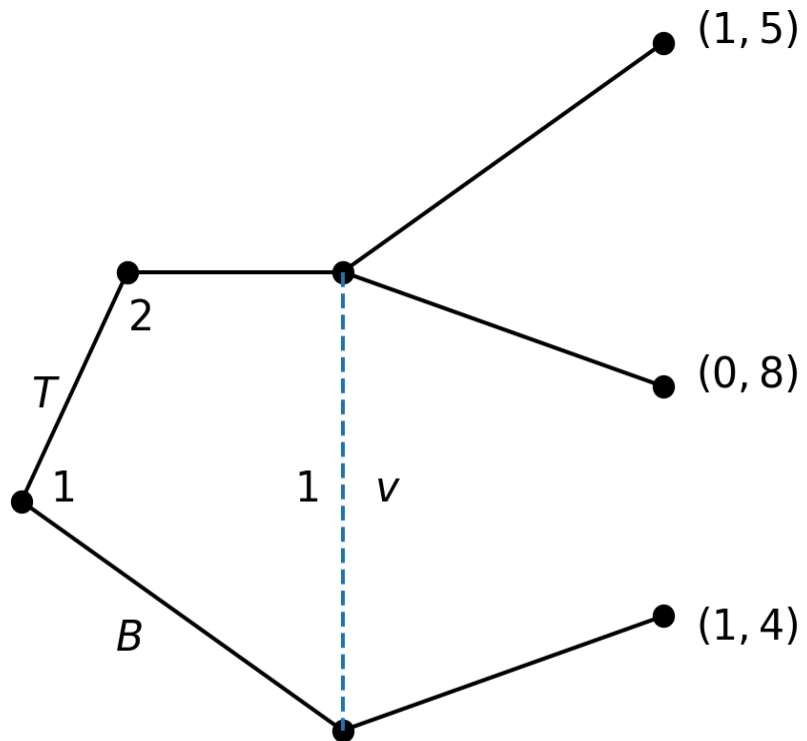
Definition 2 (Perfect recall) An extensive-form game Γ is of *perfect recall* if $\forall i \in I \setminus \{0\}, u, v \in \mathcal{U}_i, c \in \mathcal{C}_u$, and $x, y \in v, [c \gg x \iff c \gg y]$.

- Each player is allowed to remember everything he knew at previous moves and all of his choices at those moves.
- Unless explicitly indicated, henceforth all games are assumed to be of perfect recall.

Perfect recall-examples



More perfect recall-examples



Perfect or imperfect information

Definition 3 (Perfect information) If all information sets in a game are singletons, then the game is of perfect information.
If at least one information set is not a singleton, the game is of *imperfect* information.

Strategies

Definition 4 (Strategies) A **pure strategy** for player i in Γ is any map $s : \mathcal{U}_i \rightarrow \bigcup_{u \in \mathcal{U}_i} \mathcal{C}_u$ such that $s(u) \in \mathcal{C}_u \forall u \in \mathcal{U}_i$.

Let S_i be the set of player i 's pure strategies.

A **mixed strategy** for player i in Γ is any probability distribution μ over pure strategies, i.e. $\mu \in \Delta(S_i)$.

A **behavior strategy** for player i in Γ is any map $\sigma : \mathcal{U}_i \rightarrow \bigcup_{u \in \mathcal{U}_i} \Delta \mathcal{C}_u$ such that $\sigma(u) \in \Delta \mathcal{C}_u \forall u \in \mathcal{U}_i$.

Remarks:

- There is a discussion of perfect recall in Selten 1975. Selten (1975)
- Piccione and Rubinstein (1997) and Aumann, Hart, and Perry (1997) study games without perfect recall.
- Because of perfect recall, $<$ can be extended to an order between information sets $u < v, u, v \in \mathcal{U}_i$. Then, once opponents strategies are known, a player can frame her problem as a decision tree.
- Kuhn (1953): With perfect recall, behavior strategies are strategically equivalent to mixed strategies.

Common knowledge

An element of a game is *common knowledge* if every player knows it, every player knows that every player knows it, every player knows that every player knows that every player knows it, and so on ad infinitum.

- Basic structure of the game—number of players, actions available to each player, payoffs, order of moves, etc.—is known to all players.
- The structure of the game is implicitly assumed to be common knowledge.
- Assumptions about knowledge are subtle and can have strong consequences. (Example).

Example

Village of 100 sage, married couples, Myerson (1991).

- Every evening men praise their faithful wives.
- If a man obtains proof that his wife has been unfaithful, she is killed the following morning.
- If a wife is ever unfaithful, she tells all men in the village except her husband.
- Every woman in the village has been unfaithful.
- Every man knows of all infidelities except those of his own wife.

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- A holy man arrives in the village and declares: a wife in this village has been unfaithful. Does this carry any new information?

Every husband knows that 99 wives have been unfaithful. In that sense no new information is conveyed by the holy pronouncement.

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- For 99 evenings immediately following the holy pronouncement, husbands get together and continue to praise their faithful wives.
 - The 100th day, all wives are killed.

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